



Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

Dataset Overview

3,900

Transactions

Purchase records analyzed

18

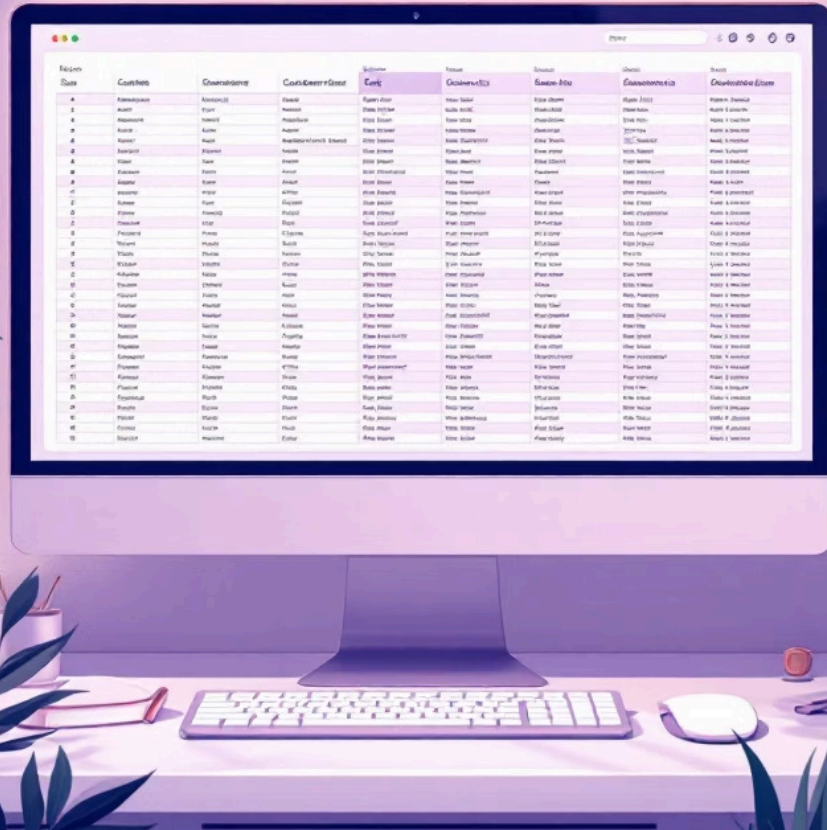
Features

Data dimensions tracked

37

Missing Values

Review ratings imputed



Index	Category	Subcategory	Product	Brand	Manufacturer	Price	Rating	Review	Location	Time	Quantity	Discount	Shipping	Payment	Order	Status	Notes
1	Electronics	Smartphones	iPhone 12	Apple	Apple Inc.	1199	4.5	Great phone	New York	2021-01-01	1	0	Free	Card	123456789	Shipped	
2	Electronics	Smartphones	Samsung Galaxy S21	Samsung	Samsung Electronics	999	4.2	Good phone	Los Angeles	2021-01-02	1	0	Free	Card	987654321	Shipped	
3	Electronics	Smartphones	Google Pixel 4	Google	Google LLC	699	4.0	Decent phone	Chicago	2021-01-03	1	0	Free	Card	456789012	Shipped	
4	Electronics	Smartphones	OnePlus 8	OnePlus	OnePlus Inc.	599	3.8	Good phone	San Francisco	2021-01-04	1	0	Free	Card	234567890	Shipped	
5	Electronics	Smartphones	Xiaomi Mi 11	Xiaomi	Xiaomi Inc.	799	4.1	Good phone	London	2021-01-05	1	0	Free	Card	345678901	Shipped	
6	Electronics	Smartphones	Huawei P50	Huawei	Huawei Technologies	899	4.3	Good phone	Paris	2021-01-06	1	0	Free	Card	567890123	Shipped	
7	Electronics	Smartphones	Motorola Edge	Motorola	Motorola Inc.	499	3.9	Good phone	Madrid	2021-01-07	1	0	Free	Card	678901234	Shipped	
8	Electronics	Smartphones	Nokia 8.3	Nokia	HMD Global	399	3.7	Good phone	Stockholm	2021-01-08	1	0	Free	Card	789012345	Shipped	
9	Electronics	Smartphones	BlackBerry Key2	BlackBerry	BlackBerry Limited	299	3.5	Good phone	Waterloo	2021-01-09	1	0	Free	Card	890123456	Shipped	
10	Electronics	Smartphones	ASUS ROG Phone 3	ASUS	ASUS Inc.	1299	4.6	Great phone	Taipei	2021-01-10	1	0	Free	Card	901234567	Shipped	
11	Electronics	Smartphones	Lenovo Legion 5	Lenovo	Lenovo Inc.	1199	4.4	Great phone	Beijing	2021-01-11	1	0	Free	Card	012345678	Shipped	
12	Electronics	Smartphones	HP Spectre x360	HP	HP Inc.	1099	4.3	Great phone	Palo Alto	2021-01-12	1	0	Free	Card	123456789	Shipped	
13	Electronics	Smartphones	Dell XPS 13	Dell	Dell Inc.	999	4.2	Great phone	Austin	2021-01-13	1	0	Free	Card	234567890	Shipped	
14	Electronics	Smartphones	Acer Swift 3	Acer	Acer Inc.	499	3.9	Great phone	Taipei	2021-01-14	1	0	Free	Card	345678901	Shipped	
15	Electronics	Smartphones	MSI Modern 15	MSI	MSI Inc.	599	3.8	Great phone	Taipei	2021-01-15	1	0	Free	Card	456789012	Shipped	
16	Electronics	Smartphones	Alienware m15	Dell	Dell Inc.	1299	4.6	Great phone	Austin	2021-01-16	1	0	Free	Card	567890123	Shipped	
17	Electronics	Smartphones	Razer Blade 15	Razer	Razer Inc.	1199	4.5	Great phone	San Jose	2021-01-17	1	0	Free	Card	678901234	Shipped	
18	Electronics	Smartphones	ASUS ZenBook 14	ASUS	ASUS Inc.	799	4.1	Great phone	Taipei	2021-01-18	1	0	Free	Card	789012345	Shipped	

Data Preparation Process

01

Data Loading

Imported using pandas

02

Exploration

Checked structure and statistics

03

Cleaning

Handled missing values

04

Standardization

Renamed to snake case

05

Feature Engineering

Created age groups and frequency

06

Database Load

Connected to PostgreSQL

Product Performance

Top-Rated Products

- Gloves: 3.86 avg rating
- Sandals: 3.84 avg rating
- Boots: 3.82 avg rating

Most Discounted Items

- Hat: 50% discounted
- Sneakers: 49.66% discounted
- Coat: 49.07% discounted



Shipping & Subscription Impact

Express Shipping

\$60.48 avg purchase

Standard Shipping

\$58.46 avg purchase

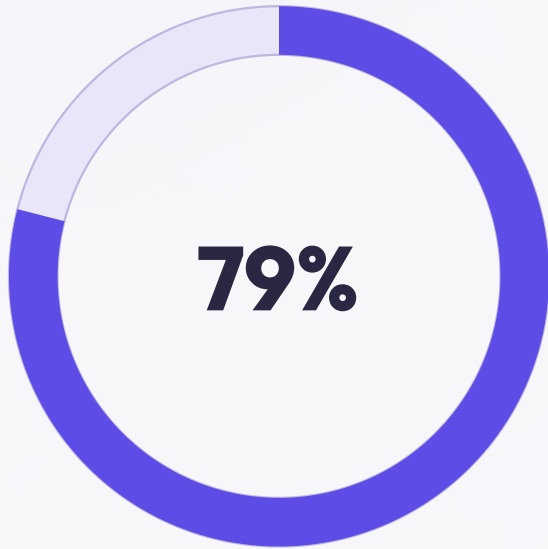
Subscribers

1,053 customers, \$59.49 avg spend

Non-Subscribers

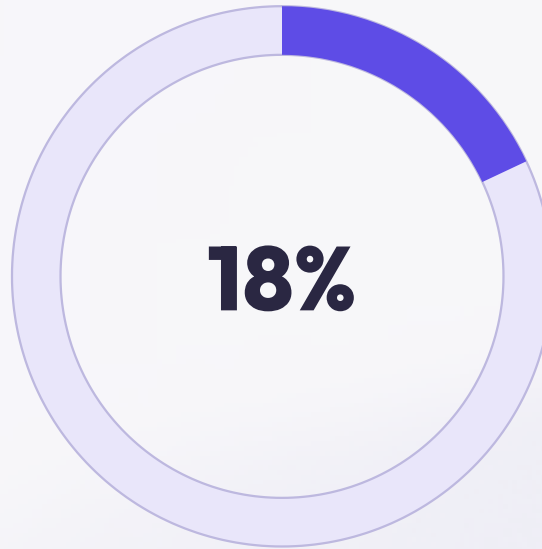
2,847 customers, \$59.87 avg spend

Customer Segmentation



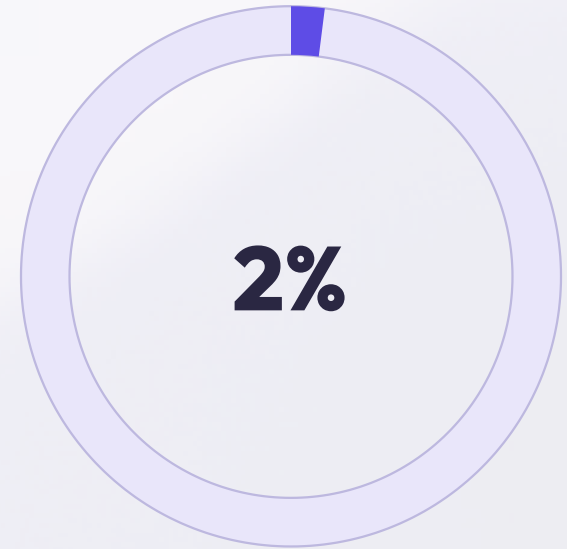
Loyal

3,116 customers



Returning

701 customers



New

83 customers

Key Insights

High-Spending Discount Users

839 customers used discounts but spent above average

Repeat Buyers Subscribe

958 repeat buyers vs. 2,518 non-repeaters

Top Products per Category

Jewelry, Blouse, Sandals, Jacket lead sales

Business Recommendations



Boost Subscriptions

Promote exclusive benefits for subscribers



Customer Loyalty Programs

Reward repeat buyers to move into "Loyal" segment



Review Discount Policy

Balance sales boosts with margin control



Targeted Marketing

Focus on high-revenue age groups and express-shipping users